Example:

```
Iptables -I INPUT 1 --dport 80 -j ACCEPT
```

It inserts a rule somewhere in a chain. The rule is inserted as the actual number that we specify.

-L: Lists all of the rules in the chain specified after the command. To list all rules in all chains in the default filter table, do not specify a chain or table. Otherwise, the following syntax should be used to list the rules in a specific chain in a particular table.

Example:

```
Iptables -L <chain-name> -t <table-name>
```

This command lists all the entries in the specified chain.

```
Iptables -L INPUT
```

-N: Creates a new chain with a user-specified name. The chain name must be unique.

Example:

```
Iptables -N allowed
```

This command tells the kernel to create a new chain of the specified name in the specified table.

-P: Sets the default policy for the specified chain, so that when packets traverse an entire chain without matching a rule, they are sent to the specified target, such as ACCEPT or DROP.

Example:

```
Iptables -P INPUT DROP
```

This command tells the kernel to set a specified default target, or policy, on a chain.

-R: Replaces a rule in the specified chain. The rule's number must be specified after the chain's name. The first rule in a chain corresponds to rule number one.

Example:

```
Iptables -R INPUT 1 -s 192.168.0.1 -j DROP
```

This command replaces the old entry at the specified line.

-X: Deletes a user-specified chain. You cannot delete a built-in chain.

Example:

```
Iptables -X allowed
```

This command deletes the specified chain from the table.

-Z: Sets the byte and packet counters in all chains for a table to zero.

Example:

```
Iptables -Z INPUT
```

This command tells the program to set all counters to zero in a specific chain, or in all chains.

Iptables parameter options

[!] -p, --protocol protocol: This is the connection protocol used. The specified protocol can be one of tcp, udp, udplite, icmp, esp, ah, sctp or the special keyword 'all'. Or it can be a numeric value, representing one of these protocols or a different one. A protocol name from */etc/protocols* is also allowed. A '!' argument before the protocol inverts the test.

[!] -s, --source address[/mask][,...]: This is the address[/mask] source specification. Address can be either a network name, a hostname, a network IP address (with /mask), or a plain IP address. A '!' argument before the address specification inverts the sense of the address.

[!] -d, --destination address[/mask][,...]: This is the address[/mask] destination specification.

-j, --jump target: This jumps to the specified target when a packet matches a particular rule.

[!] -i, --in-interface name: This is the name of an interface via which a packet has been received (only for packets entering the INPUT, FORWARD and PREROUTING chains). When the '!' argument is used before the interface name, the sense is inverted. If the interface name ends in a '+', then any interface which begins with this name will match. If this option is omitted, any interface name will match.

[!] -o, --out-interface name: This is the name of an interface via which a packet is going to be sent (for packets entering the FORWARD, OUTPUT and POSTROUTING chains). When the '!' argument is used before the interface name, the sense is inverted. If the interface name ends in a '+', then any interface which begins with this name will match. If this option is omitted, any interface name will match.

[!] -f, --fragment: This rule applies only to fragmented packets.

You can use the exclamation point character (!) option before this parameter to specify that only unfragmented packets are matched.

-c, --set-counters packets bytes: This enables the administrator to initialise the packet and byte counters of a rule (during the INSERT, APPEND and REPLACE operations).

Writing a simple rule set of Iptables

The Iptables' rule with respect to allowing loopback connections is as follows:

```
sudo Iptables -A INPUT -i lo -j ACCEPT
sudo Iptables -A OUTPUT -o lo -j ACCEPT
```